

WEEK6 PRACTICAL:

Task1:

Create a client-server application using Named Pipes (FIFOs). The client sends messages and the server processes and responds to them. Analyze FIFO behavior when multiple clients communicate simultaneously.

```
Last login: Thu Sep 17 09:07:48 on console
(eval):8: command not found: >
vishalkarnati@VISHALS-MacBook-Air ~ % cd ~/Desktop/os
vishalkarnati@VISHALS-MacBook-Air os % mkdir week6pract
vishalkarnati@VISHALS-MacBook-Air os % cd week6pract
vishalkarnati@VISHALS-MacBook-Air week6pract % nano server.c
vishalkarnati@VISHALS-MacBook-Air week6pract %
```

```
UW PICO 5.09 File: server.c Modified
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/stat.h>

#define REQUEST_FIFO "request_fifo"

int main()
{
    char message[200];
    char response[300];

    mkfifo(REQUEST_FIFO, 0660);

    printf("Server started...\n");
    printf("Waiting for clients...\n");

    int fd = open(REQUEST_FIFO, O_RDONLY);

    while (1)
    {
        int n = read(fd, message, sizeof(message) - 1);

        if (n > 0)
        {
            message[n] = '\0';

            printf("Server received: %s\n", message);

            char client_fifo[100];
            char client_id[60];

            sscanf(message, "%s %[^\n]", client_id, message);
            snprintf(client_fifo, sizeof(client_fifo),
                    "response_%s", client_id);

            snprintf(response, sizeof(response),
                    "Server processed your message: %s", message);

            int response_fd = open(client_fifo, O_WRONLY);
            write(response_fd, response, strlen(response) + 1);
            close(response_fd);
        }
    }

    close(fd);

    return 0;
}
```

```
vishalkarnati@VISHALS-MacBook-Air week6pract % nano client.c
vishalkarnati@VISHALS-MacBook-Air week6pract %
```

```
UW PIC0 5.09 File: client.c Modified
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/stat.h>

#define REQUEST_FIFO "request_fifo"

int main(int argc, char *argv[])
{
    if (argc < 3)
    {
        printf("Usage: ./client <client_id> <message>\n");
        return 1;
    }

    char *client_id = argv[1];
    char *message = argv[2];

    char client_fifo[100];
    char request[300];
    char response[300];

    snprintf(client_fifo, sizeof(client_fifo),
             "response_%s", client_id);

    mkfifo(client_fifo, 0666);

    snprintf(request, sizeof(request),
             "%s %s", client_id, message);

    int request_fd = open(REQUEST_FIFO, O_WRONLY);
    write(request_fd, request, strlen(request) + 1);
    close(request_fd);

    int response_fd = open(client_fifo, O_RDONLY);

    int n = read(response_fd, response, sizeof(response) - 1);

    if (n > 0)
    {
        response[n] = '\0';
        printf("Client %s received: %s\n", client_id, response);
    }

    close(response_fd);

    unlink(client_fifo);

    return 0;
}
```

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```
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(eval):8: command not found: >
vishalkarnati@VISHALS-MacBook-Air ~ % cd ~/Desktop/os
vishalkarnati@VISHALS-MacBook-Air os % mkdir week6pract
vishalkarnati@VISHALS-MacBook-Air os % cd week6pract
vishalkarnati@VISHALS-MacBook-Air week6pract % nano server.c
vishalkarnati@VISHALS-MacBook-Air week6pract % nano client.c
vishalkarnati@VISHALS-MacBook-Air week6pract % gcc server.c -o server
vishalkarnati@VISHALS-MacBook-Air week6pract % gcc client.c -o client
vishalkarnati@VISHALS-MacBook-Air week6pract % ./server
Server started...
Waiting for clients...
request_fifo
Server received: 1 Hello
Server received: 1 Hello
Server received: 2 HowAreYou
Server received: 3 GoodMorning
^C
vishalkarnati@VISHALS-MacBook-Air week6pract %
```

```
Last login: Thu Sep 17 13:22:02 on ttys001
(eval):8: command not found: >
vishalkarnati@VISHALS-MacBook-Air ~ % cd ~/Desktop/os/week6pract
vishalkarnati@VISHALS-MacBook-Air week6pract % ./client 1 Hello
Client 1 received: Server processed your message: Hello
vishalkarnati@VISHALS-MacBook-Air week6pract % ./client 2 HowAreYou
Client 2 received: Server processed your message: HowAreYou
vishalkarnati@VISHALS-MacBook-Air week6pract % ./client 3 GoodMorning
Client 3 received: Server processed your message: GoodMorning
vishalkarnati@VISHALS-MacBook-Air week6pract %
```

Task2:

Create a POSIX signal handling program that captures SIGINT, SIGTERM, and SIGUSR1. Demonstrate asynchronous event handling and explain the role of signal handlers.

```
UW PICO 5.09 File: signals.c Modified
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <signal.h>

volatile sig_atomic_t sigint_received = 0;
volatile sig_atomic_t sigterm_received = 0;
volatile sig_atomic_t sigusr1_received = 0;

void handle_signal(int signal)
{
    if (signal == SIGINT)
    {
        sigint_received = 1;
    }
    else if (signal == SIGTERM)
    {
        sigterm_received = 1;
    }
    else if (signal == SIGUSR1)
    {
        sigusr1_received = 1;
    }
}

int main()
{
    struct sigaction sa;

    sa.sa_handler = handle_signal;
    sigemptyset(&sa.sa_mask);
    sa.sa_flags = 0;

    sigaction(SIGINT, &sa, NULL);
    sigaction(SIGTERM, &sa, NULL);
    sigaction(SIGUSR1, &sa, NULL);

    printf("Process started.\n");
    printf("PID = %d\n", getpid());
    printf("Waiting for signals...\n");

    while (1)
    {
        sleep(1);

        if (sigint_received)
        {
            printf("SIGINT received\n");
            sigint_received = 0;
        }

        if (sigusr1_received)
        {
            printf("SIGUSR1 received\n");
            sigusr1_received = 0;
        }

        if (sigterm_received)
        {
            printf("SIGTERM received\n");
            sigterm_received = 0;
        }
    }
}
```

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```
UW PICO 5.09 File: signals.c Modified
sigemptyset(&sa.sa_mask);
sa.sa_flags = 0;

sigaction(SIGINT, &sa, NULL);
sigaction(SIGTERM, &sa, NULL);
sigaction(SIGUSR1, &sa, NULL);

printf("Process started.\n");
printf("PID = %d\n", getpid());
printf("Waiting for signals...\n");

while (1)
{
    sleep(1);

    if (sigint_received)
    {
        printf("SIGINT received\n");
        sigint_received = 0;
    }

    if (sigusr1_received)
    {
        printf("SIGUSR1 received\n");
        sigusr1_received = 0;
    }

    if (sigterm_received)
    {
        printf("SIGTERM received\n");
        printf("Terminating program...\n");
        break;
    }
}

return 0;
}
```

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```
~/Desktop/os/week6pract -- -zsh
Last login: Thu Sep 17 13:24:00 on ttys001
(qval):8: command not found: >
vishalkarnati@VISHALS-MacBook-Air ~ % cd ~/Desktop/os/week6pract
vishalkarnati@VISHALS-MacBook-Air week6pract % nano signals.c
vishalkarnati@VISHALS-MacBook-Air week6pract % gcc signals.c -o signals
vishalkarnati@VISHALS-MacBook-Air week6pract % ./signals
Process started.
PID = 2109
Waiting for signals...
SIGUSR1 received
SIGINT received
SIGTERM received
Terminating program...
vishalkarnati@VISHALS-MacBook-Air week6pract %
```

```
~/Desktop/os/week6pract -- -zsh
Last login: Thu Sep 17 13:29:06 on ttys000
(qval):8: command not found: >
vishalkarnati@VISHALS-MacBook-Air ~ % cd ~/Desktop/os/week6pract
vishalkarnati@VISHALS-MacBook-Air week6pract % kill -SIGUSR1 2109
vishalkarnati@VISHALS-MacBook-Air week6pract % kill -SIGINT 2109
vishalkarnati@VISHALS-MacBook-Air week6pract % kill -SIGTERM 2109
vishalkarnati@VISHALS-MacBook-Air week6pract %
```